

Computer Science - Undergraduate Level Assessment 3

Instructions

- Answer all questions.

Multiple Choice

Question 1

(2 marks)

Which of the following is true of interrupts?

- A. They are generated when memory cycles are "stolen".
- B. They are used in place of data channels.
- C. They can indicate completion of an I/O operation.
- D. They cannot be generated by arithmetic operations.

Question 2

(2 marks)

Church's thesis equates the concept of "computable function" with those functions computable by, for example, Turing machines. Which of the following is true of Church's thesis?

- A. It was first proven by Alan Turing.
- B. It has not yet been proven, but finding a proof is a subject of active research.
- C. It can never be proven.
- D. It is now in doubt because of the advent of parallel computers.

Question 3

(2 marks)

Which of the following has the greatest potential for compromising a user's personal privacy?

- A. A group of cookies stored by the user's Web browser
- B. The Internet Protocol (IP) address of the user's computer

C. The user's e-mail address

D. The user's public key used for encryption

Question 4

(2 marks)

Let `l = [1,2,3,4]`. What is `min(l)` in Python 3?

A. 1

B. 2

C. 3

D. 4

Question 5

(2 marks)

Which of the following pairs of 8-bit, two's-complement numbers will result in overflow when the members of the pairs are added?

A. 11111111, 00000001

B. 00000001, 10000000

C. 11111111, 10000001

D. 10000001, 10101010

True / False

Question 1

(2 marks)

The expression `print(list[1:3])` in Python 3 outputs the entire list `['abcd', 786, 2.23, 'john', 70.2]`.

True

False

Question 2

(2 marks)

The value of the Python expression $1 + 3 * 3$ is 4, since the modulo operation is calculated after addition.
The value of the Python expression $1 + 3 \% 3$ is 4, since the modulo operation is calculated after addition.

☐ True

☐ False

Question 3

(2 marks)

Recursive procedures require stack-based storage allocation to manage multiple instances of function calls.

☐ True

☐ False

Question 4

(2 marks)

The language $\{ww \mid w \text{ in } (0 + 1)^*\}$ is accepted by some Turing machine, but it cannot be accepted by any pushdown automaton.

☐ True

☐ False

Question 5

(2 marks)

In Python 3, the function `max(list)` returns the item from the list with the maximum value.

☐ True

☐ False

Long Form

Question 1

(10 marks)

Write a function to calculate the area of a tetrahedron.

Type your answer here...



Question 2

(10 marks)

Write a function to split the given string at uppercase letters by using regex.

Type your answer here...



Submit Exam